

ANSWER KEY — Integration Lesson 2 Test: Area Under a Graph = the Total

For the grader · full working shown · give credit for correct method with small arithmetic slips

1. A hose fills a barrel at a steady 7 liters per minute for 6 minutes. What is the area of the rectangle under the flow-rate graph, and how much water is in the barrel?

Work: rectangle: width 6 min, height 7 L/min $\rightarrow$ area = $7 \times 6 = 42$. Units: (L/min) $\times$ min = L. Area under a rate graph = the total. **Answer: area = 42, meaning 42 liters of water.**

2. A van drives at a steady 45 miles per hour from $t = 2$ hours to $t = 6$ hours. How far does it travel during that stretch?

Work: width = $6 - 2 = 4$ hours (NOT 6). Distance = $45 \times 4 = 180$. **Answer: 180 miles.** (Classic error: using width 6 $\rightarrow$ 270 miles, wrong — the stretch starts at $t = 2$, not $t = 0$.)

3. A bicycle starts from rest and speeds up steadily to 12 m/s over 6 seconds. How far does it travel?

Work: slanted line from 0 $\rightarrow$ triangle. Area = $\frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 6 \times 12 = 36$. **Answer: 36 meters.** (Classic error: $12 \times 6 = 72$ forgets the $\frac{1}{2}$ — the bike was only at 12 m/s for the last instant.)

4. A ball is dropped; its speed grows steadily from 0 to 20 m/s during the first 2 seconds. How far does it fall?

Work: triangle: base 2 s, height 20 m/s $\rightarrow \frac{1}{2} \times 2 \times 20 = 20$. **Answer: 20 meters.**

5. A scooter already moving at 5 m/s speeds up steadily to 15 m/s over 4 seconds. Find the distance via rectangle + triangle.

Work: rectangle (starting speed held the whole time): $5 \times 4 = 20$ m. Triangle (extra speed grows 0 $\rightarrow$ $15 - 5 = 10$): $\frac{1}{2} \times 4 \times 10 = 20$ m. Total = $20 + 20 = 40$. **Answer: 40 meters.** (Watch for triangle height 15 instead of 10 — the triangle only covers the EXTRA speed above 5.)

6. Rainfall rate steady at 3 mm/hour for 5 hours. Find the area, its unit, and its meaning.

Work: area = $3 \times 5 = 15$. Unit: (mm/hour) $\times$ hours = mm — the hours cancel. Meaning: the total depth of rain that fell, i.e. what a rain gauge would read. **Answer: 15 millimeters of total rainfall.**

7. A race car accelerates from rest with a straight-line speed graph and covers 48 meters in 8 seconds. What top speed does it reach?

Work: triangle with known area, unknown height: $\frac{1}{2} \times 8 \times h = 48 \rightarrow 4h = 48 \rightarrow h = 12$. Check: $\frac{1}{2} \times 8 \times 12 = 48$ ✓. **Answer: 12 m/s.**

8. Earning rate climbs steadily from \$10/day to \$20/day over 5 days. Total earned?

Work: rectangle: $10 \times 5 = \$50$. Triangle: extra rate grows $0 \rightarrow (20 - 10) = 10$, so $\frac{1}{2} \times 5 \times 10 = \25 . Total = $50 + 25 = 75$. **Answer: \$75.**

9. A car traveling 24 m/s brakes steadily to a stop over 4 seconds. How far does it travel while stopping?

Work: downhill slanted line from 24 to 0 $\rightarrow$ still a triangle, base 4, height 24 (height at the start). Distance = $\frac{1}{2} \times 4 \times 24 = 48$. **Answer: 48 meters.** (Common mistake: thinking a braking car covers no distance — it moves the whole time it slows.)

10. Tram: 8 m/s for 5 s, then steadily 8 $\rightarrow$ 16 m/s over 4 s, then 16 m/s for 10 s. Total distance?

Work: piece 1 (rectangle): $8 \times 5 = 40$ m. Piece 2 (trapezoid = rectangle + triangle): $8 \times 4 = 32$ m plus $\frac{1}{2} \times 4 \times (16 - 8) = 16$ m $\rightarrow 48$ m. Piece 3 (rectangle): $16 \times 10 = 160$ m. Total = $40 + 48 + 160 = 248$. **Answer: 248 meters.**